Tutor Meeting Brief (EN) — Final Test/Holdout Results

One-line conclusion: the Wave-1 validation champion `gpt2_miflu` (0.771) did **not** generalize to the held-out COVID-era Test window — there the simple linear model `nlinear` (2.739) significantly beats it (3.051, $p = 4.0 \times 10^{-4}$); both models still beat the zero-parameter `seasonal_naive` baseline (4.279), which collapses under distribution shift.

Three facts the tutor must see

Fact 1: the validation ranking reverses on Test

model	validation avg MSE 4L	Test avg MSE 4L	95% CI (Test)
`nlinear`	1.070	**2.739**	[2.731, 2.747]
`gpt2_miflu`	**0.771**	3.051	[2.922, 3.179]
`seasonal_naive`	0.771	4.279	n/a

The champion's validation advantage was window-specific. On Test, `nlinear` is better at **all four horizons** (L24 +0.413, $p = 0.004$; L36 +0.477, $p = 0.004$; L48 $p = 0.055$; L60 $p = 0.175$). Confirmatory test C1 (paired $t = 5.46$, $p = 4.0 \times 10^{-4}$, Holm-corrected) rejects the hypothesis that the two are equal.

Fact 2: seasonality is not a forecast — the baseline collapses on Test

	validation	Test	gap
`seasonal_naive`	0.771	4.279	**+3.51**

On validation, `seasonal_naive` was tied with `gpt2_miflu` — its success was memorizing the strongly seasonal 2016–2018 period. On the COVID-era Test window it is the worst model by a wide margin, and its peak-intensity error reaches 124%. This is the study's cleanest demonstration that "repeat last year" is not forecasting.

Fact 3: `nlinear` is the best operational model on Test (and the simplest)

	params	Test avg MSE 4L	Test PIE (peak intensity)
`nlinear`	2,520	2.739	9–21% (hit 75–94%)
`gpt2_miflu`	1,030,056	3.051	41–62%

For peak intensity forecasting — the metric that matters for outbreak detection — `nlinear` is dramatically better. Both learned models still beat the baseline (C2 $p = 4.6 \times 10^{-9}$, C3 $p = 7.7 \times 10^{-21}$).

Questions to discuss / decide with the tutor

1. **Framing the final narrative**: should the paper's headline be "a simple linear forecaster outperforms an LLM on held-out data" rather than any LLM-superiority claim?
2. **Champion status**: the frozen Wave-1 selection stands as a validation result, but the Test winner is `nlinear`. How do we phrase "frozen selection" vs "final recommendation" so neither contradicts the other?
3. **`seasonal_naive` as a formal baseline**: its validation-vs-Test collapse is the most illustrative result — promote it to a main-table baseline?
4. **Cost argument**: `nlinear` is 409× fewer params and ~28× faster to train, and now also more accurate on Test. Does the LLM machinery retain any justification in the final

paper?

5. **Limitations wording**: the Test window is a single contiguous COVID-era window — confirm we state "distribution shift conflated with COVID" explicitly rather than over-claiming robustness.

Quick numbers (citable)

- Test primary: `nlinear` 2.739, `gpt2_miflu` 3.051, `seasonal_naive` 4.279.
- Confirmatory (all Holm-significant, m = 3): C1 gpt2 vs nlinear p = 4.0×10^{-4}; C2 gpt2 vs seasonal p = 4.6×10^{-9}; C3 nlinear vs seasonal p = 7.7×10^{-21}.
- Validation-to-Test gap: nlinear +1.67, gpt2 +2.28, seasonal +3.51.
- Test peak intensity (PIE): nlinear 9–21%, gpt2 41–62%, seasonal 50–124%.
- Integrity: 84/84 runs, 25/25 terminal audit, no 85th job, frozen script hash unchanged.

Artifact paths: full reports `MIFlu_Final_Supervisor_Report_EN.md` / `_ZH.md` (+ .docx/.pdf); metrics and figures under `results/v3_clean/final/`.